

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		Substitution: $\frac{1}{R} = \frac{1}{12} + \frac{1}{6}$ (1) $\frac{1}{R} = \frac{1}{4}$ or 0.25 (1) accept 0.24 if decimals are used $R = 4 \text{ } [\Omega]$ (1) accept 4.2 $[\Omega]$	1	1 1		3	3	3
		(ii)		$6 - 4 \text{ ecf} = 2 \text{ } [\Omega]$		1		1		1
		(iii)		Substitution : $I = \frac{9}{6}$ (1) $= 1.5 \text{ [A]} (1)$	1	1		2	2	2
		(iv)		Substitution: $1.5 \text{ ecf} = \frac{V}{2 \text{ ecf}}$ (1) Rearrangement: $V = 1.5 \times 2$ (1) $= 3 \text{ [V]} (1)$ Alternative: $2 \text{ } \Omega = \frac{1}{3}$ of the total resistance (6 Ω) (1) So $V = \frac{1}{3}$ of 9 V (1) $= 3 \text{ [V]} (1)$	1	1 1		3	3	3

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	(b)			Substitution: $P = VI = 9 \times 4.5$ or $P = I^2R = 4.5^2 \times 2$ ecf from (a)(ii) (1) $= 40.5$ [W] (1) Either for the 3rd mark: In 10 s, $E = P \times t = 40.5 \times 10 = 405$ [J] so disagree Or: $P = \frac{E}{t} = \frac{45}{10} = 4.5$ [W] so disagree Alternative: $P = \frac{45}{10}$ (1) $= 4.5$ [W] (1) Either for the 3rd mark: $I = \frac{P}{V} = \frac{4.5}{9} = 0.5$ [A] so disagree Or: $V = \frac{P}{I} = \frac{4.5}{4.5} = 1$ [V] so disagree Conclusion must be present to award 3 marks			3	3	3	3
				Question 7 total	3	6	3	12	11	12